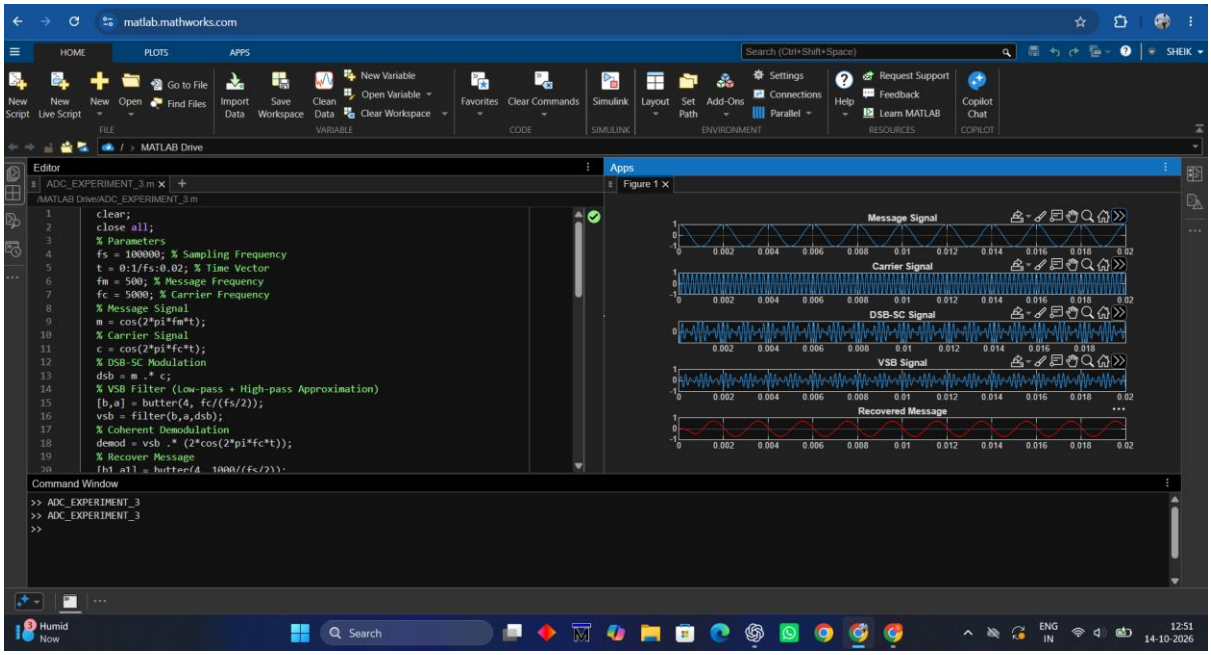
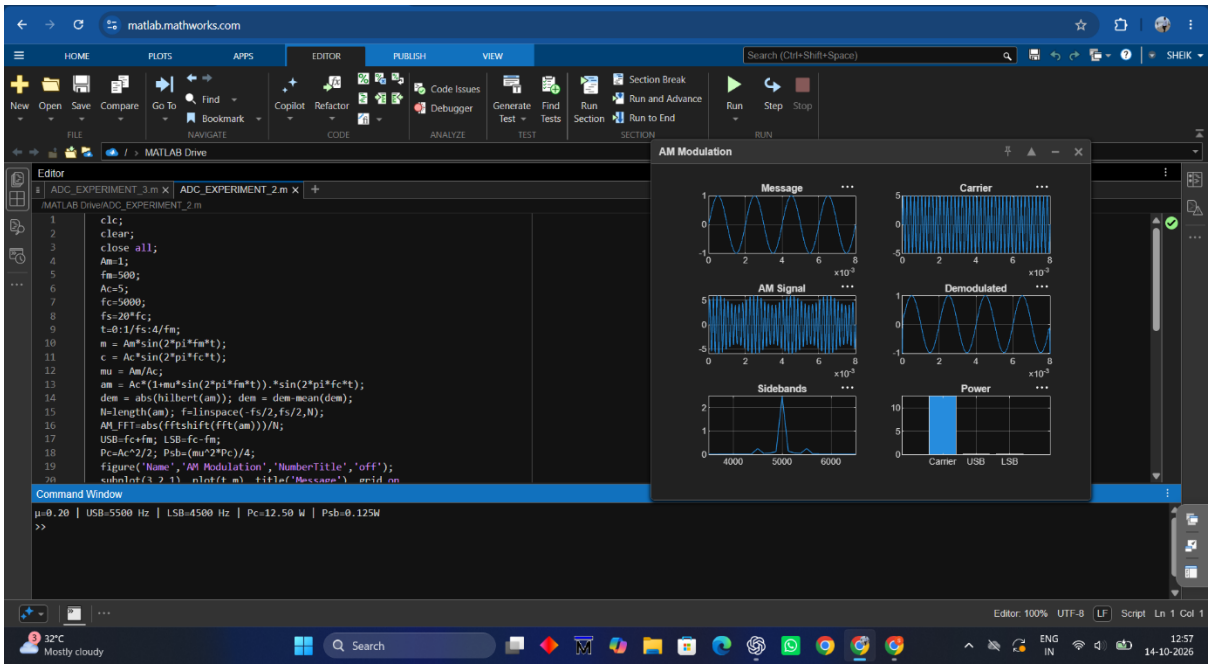
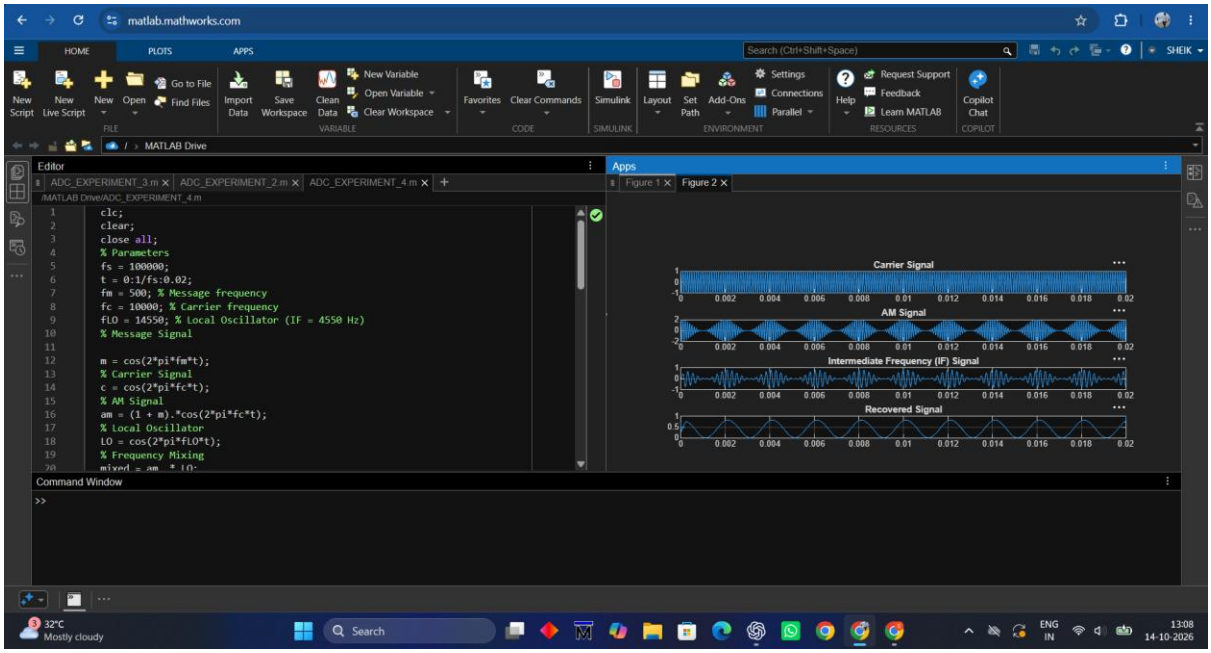
EXP 3


matlab.mathworks.com
HOME
PLOTS
APPS
MATLAB Drive
Editor
ADC_EXPERIMENT_3.m
Apps
Figure 1
Message Signal
Carrier Signal
DSB-SC Signal
VSB Signal
Recovered Message
Command Window
>> ADC_EXPERIMENT_3
>> ADC_EXPERIMENT_3

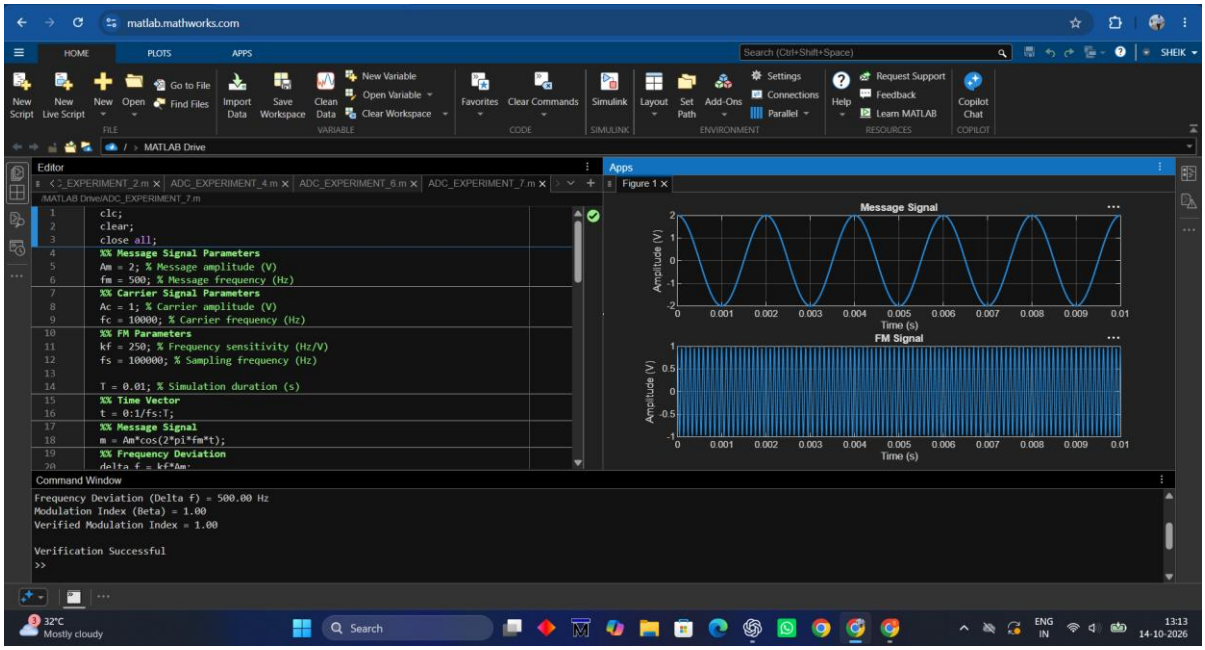
EXP 2


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HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
MATLAB Drive
Editor
ADC_EXPERIMENT_3.m
ADC_EXPERIMENT_2.m
AM Modulation
Message
Carrier
AM Signal
Demodulated
Sidebands
Power
Command Window
μ=0.20 | USB=5500 Hz | LSB=4500 Hz | Pc=12.50 W | Psb=0.125W

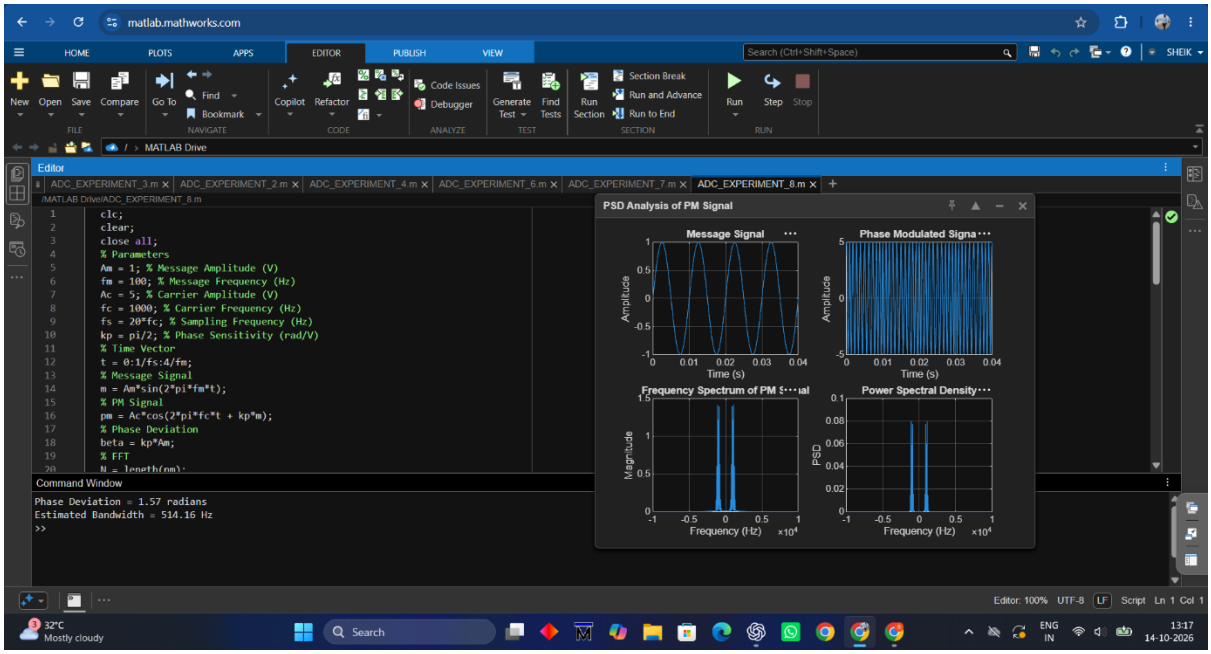
EXP 4


matlab.mathworks.com
HOME
PLOTS
APPS
MATLAB Drive
Editor
ADC_EXPERIMENT_4.m
clc;
clear;
close all;
% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500; % Message frequency
fc = 10000; % Carrier frequency
fLO = 14550; % Local Oscillator (IF = 4550 Hz)
% Message Signal
m = cos(2*pi*fm*t);
% Carrier Signal
c = cos(2*pi*fc*t);
% AM Signal
am = (1 + m).*cos(2*pi*fc*t);
% Local Oscillator
LO = cos(2*pi*fLO*t);
% Frequency Mixing
Apps
Figure 1
Figure 2
Carrier Signal
AM Signal
Intermediate Frequency (IF) Signal
Recovered Signal
Command Window
>>
32°C
Mostly cloudy
Search
ENG
IN
13:08
14-10-2026

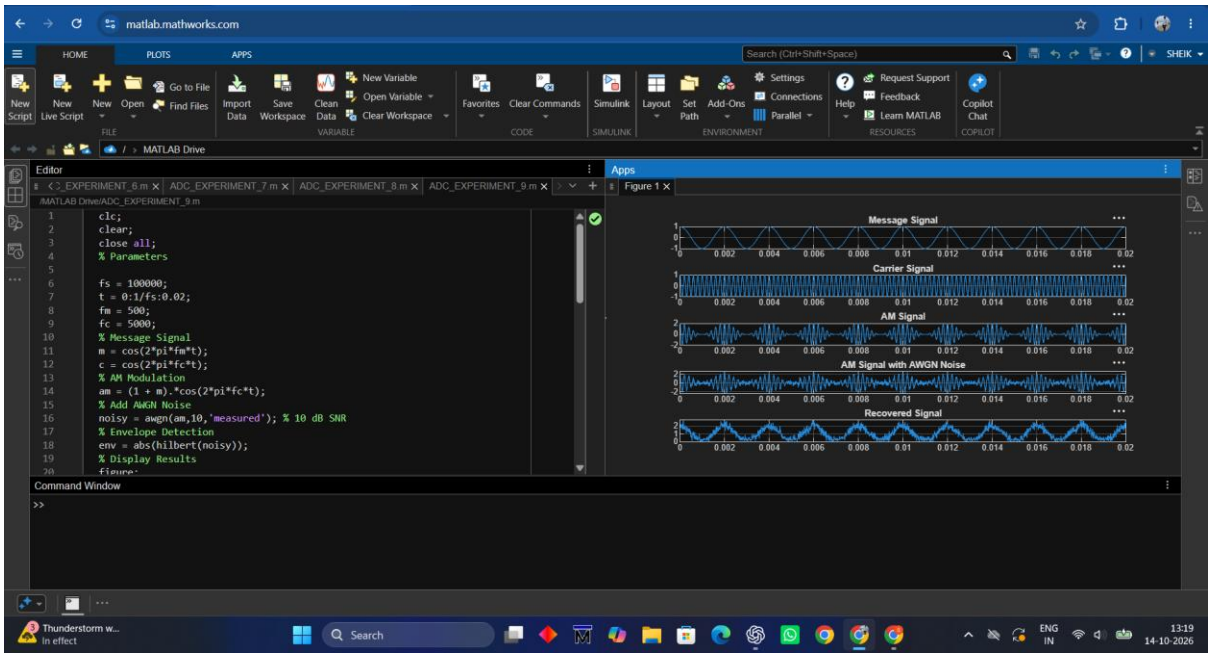
EXP 7


matlab.mathworks.com
HOME
PLOTS
APPS
MATLAB Drive
Editor
ADC_EXPERIMENT_7.m
clc;
clear;
close all;
%% Message Signal Parameters
Am = 2; % Message amplitude (V)
fm = 500; % Message frequency (Hz)
%% Carrier Signal Parameters
Ac = 1; % Carrier amplitude (V)
fc = 10000; % Carrier frequency (Hz)
%% FM Parameters
kf = 250; % Frequency sensitivity (Hz/V)
fs = 100000; % Sampling frequency (Hz)
T = 0.01; % Simulation duration (s)
%% Time Vector
t = 0:1/fs:T;
%% Message Signal
m = Am*cos(2*pi*fm*t);
%% Frequency Deviation
Apps
Figure 1
Message Signal
FM Signal
Amplitude (V)
Time (s)
Command Window
Frequency Deviation (Delta f) = 500.00 Hz
Modulation Index (Beta) = 1.00
Verified Modulation Index = 1.00
Verification Successful
>>
32°C
Mostly cloudy
Search
ENG
IN
13:13
14-10-2026

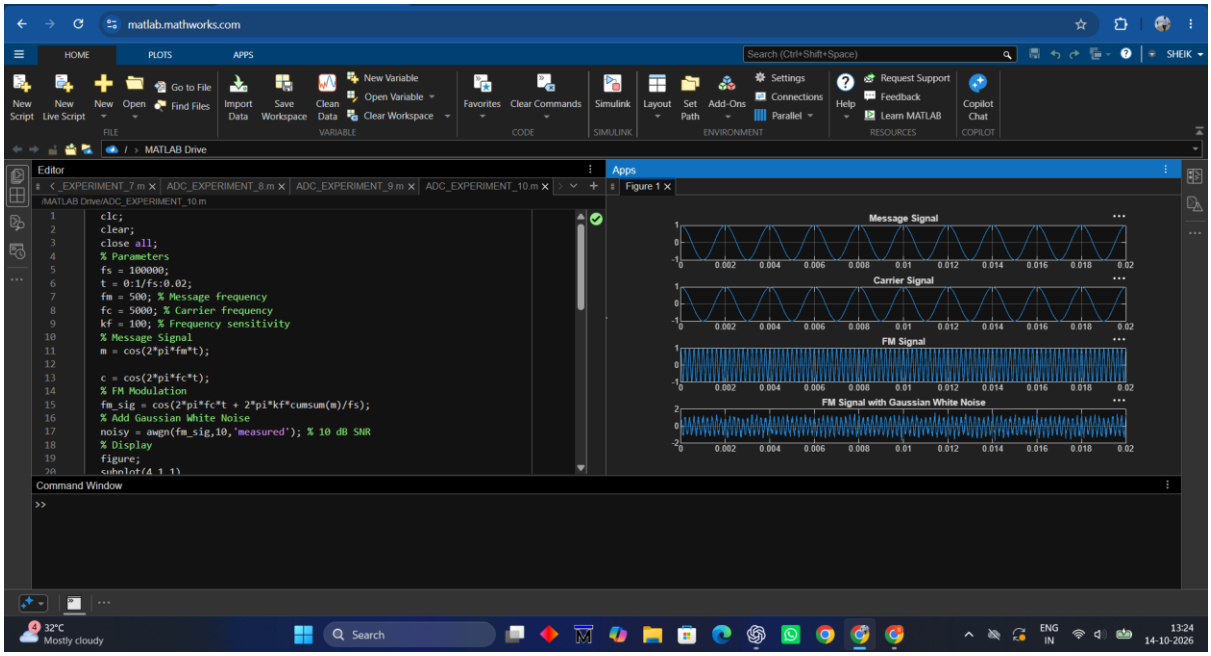
EXP 8


matlab.mathworks.com
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
MATLAB Drive
ADC_EXPERIMENT_3.m
ADC_EXPERIMENT_2.m
ADC_EXPERIMENT_4.m
ADC_EXPERIMENT_6.m
ADC_EXPERIMENT_7.m
ADC_EXPERIMENT_8.m
clc;
clear;
close all;
% Parameters
Am = 1; % Message Amplitude (V)
fm = 100; % Message Frequency (Hz)
Ac = 5; % Carrier Amplitude (V)
fc = 1000; % Carrier Frequency (Hz)
fs = 20*fc; % Sampling Frequency (Hz)
kp = pi/2; % Phase Sensitivity (rad/V)
% Time Vector
t = 0:1/fs:4/fm;
% Message Signal
m = Am*sin(2*pi*fm*t);
% PM Signal
pm = Ac*cos(2*pi*fc*t + kp*m);
% Phase Deviation
beta = kp*Am;
% FFT
PSD Analysis of PM Signal
Message Signal
Phase Modulated Signa
Frequency Spectrum of PM S
Power Spectral Density
Amplitude
Time (s)
Magnitude
PSD
Frequency (Hz)
Command Window
Phase Deviation = 1.57 radians
Estimated Bandwidth = 514.16 Hz
Editor: 100%
UTF-8
Script
Ln 1 Col 1
32°C
Mostly cloudy
Search
13:17
14-10-2026

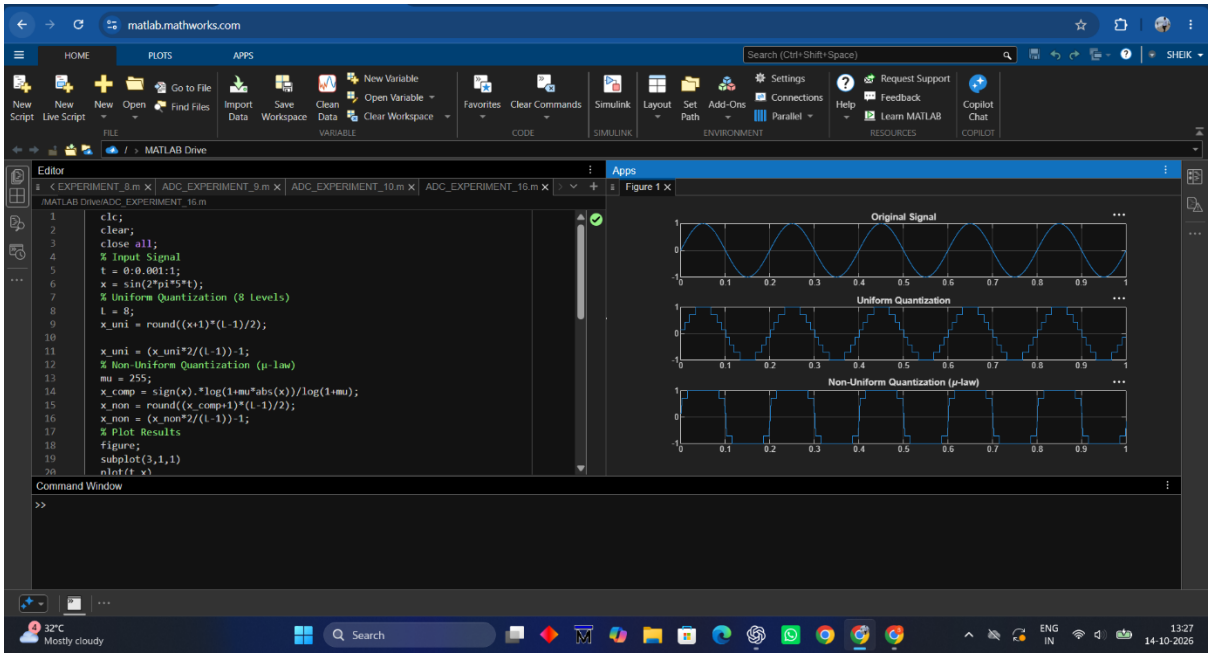
EXP 9


matlab.mathworks.com
HOME
PLOTS
APPS
MATLAB Drive
Editor
Apps
Figure 1
ADC_EXPERIMENT_6.m
ADC_EXPERIMENT_7.m
ADC_EXPERIMENT_8.m
ADC_EXPERIMENT_9.m
clc;
clear;
close all;
% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;
fc = 5000;
% Message Signal
m = cos(2*pi*fm*t);
c = cos(2*pi*fc*t);
% AM Modulation
am = (1 + m).*cos(2*pi*fc*t);
% Add AWGN Noise
noisy = awgn(am,10,'measured'); % 10 dB SNR
% Envelope Detection
env = abs(hilbert(noisy));
% Display Results
Message Signal
Carrier Signal
AM Signal
AM Signal with AWGN Noise
Recovered Signal
Command Window
Thunderstorm w...
In effect
Search
13:19
14-10-2026

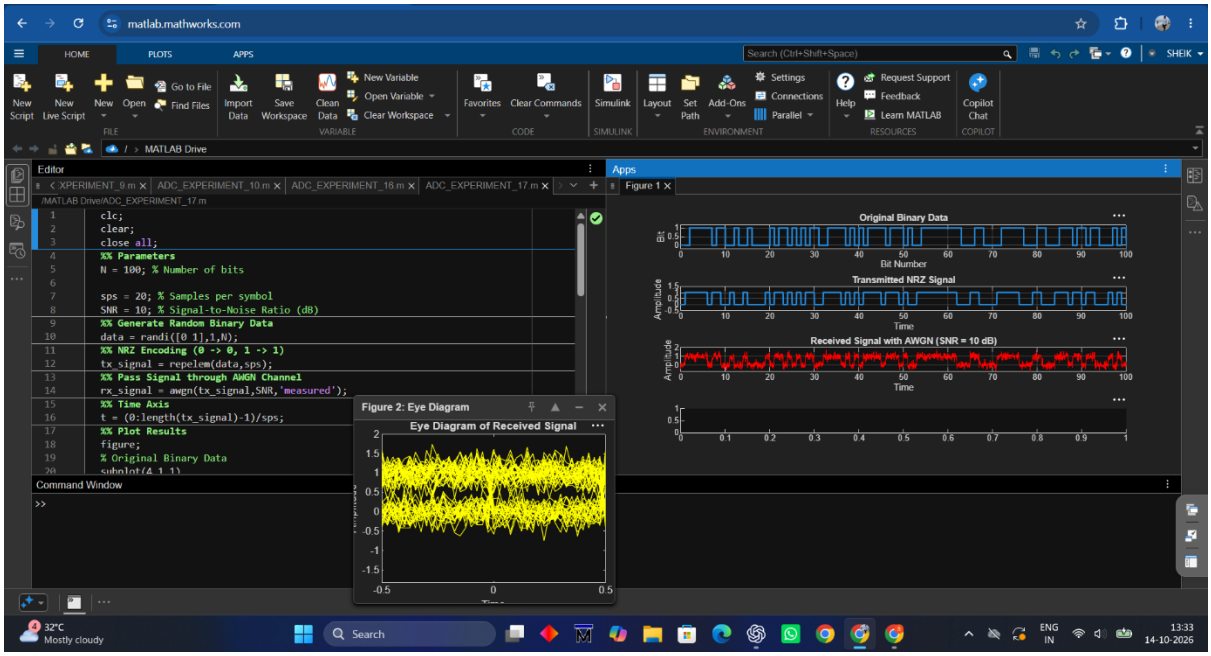
EXP 10


matlab.mathworks.com
HOME
PLOTS
APPS
Search (Ctrl+Shift+Space)
SHEIK
New Script
New Live Script
New
Open
Go to File
Find Files
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
Favorites
Clear Commands
Simulink
Layout
Set Path
Add-Ons
Settings
Connections
Parallel
Help
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Learn MATLAB
Copilot Chat
FILE
VARIABLE
CODE
SIMULINK
ENVIRONMENT
RESOURCES
COPILOT
MATLAB Drive
Editor
_EXPERIMENT_7.m
ADC_EXPERIMENT_8.m
ADC_EXPERIMENT_9.m
ADC_EXPERIMENT_10.m
/MATLAB Drive/ADC_EXPERIMENT_10.m
clc;
clear;
close all;
% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500; % Message frequency
fc = 5000; % Carrier frequency
kf = 100; % Frequency sensitivity
% Message Signal
m = cos(2*pi*fm*t);
c = cos(2*pi*fc*t);
% FM Modulation
fm_sig = cos(2*pi*fc*t + 2*pi*kf*cumsum(m)/fs);
% Add Gaussian White Noise
noisy = awgn(fm_sig,10,'measured'); % 10 dB SNR
% Display
figure;
subplot(4,1,1)
Apps
Figure 1
Message Signal
Carrier Signal
FM Signal
FM Signal with Gaussian White Noise
Command Window
>>
32°C
Mostly cloudy
Search
ENG
IN
13:24
14-10-2026

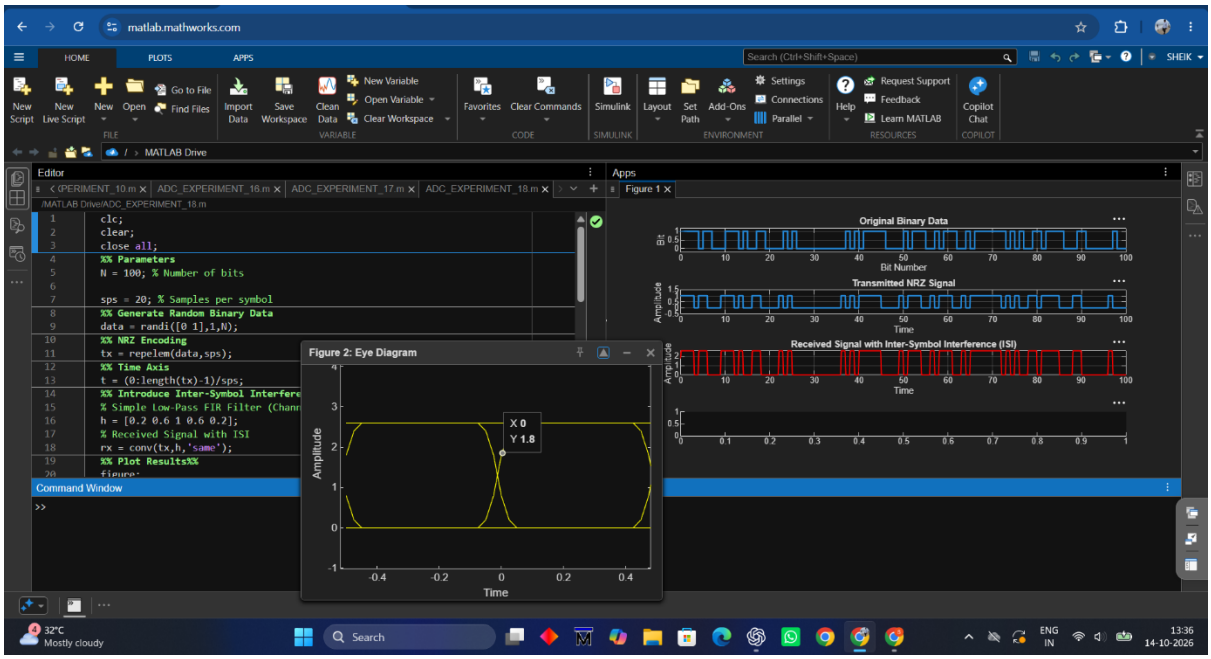
EXP 16


matlab.mathworks.com
HOME
PLOTS
APPS
Search (Ctrl+Shift+Space)
SHEIK
New Script
New Live Script
New
Open
Go to File
Find Files
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
Favorites
Clear Commands
Simulink
Layout
Set Path
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ENVIRONMENT
RESOURCES
COPILOT
MATLAB Drive
Editor
EXPERIMENT_8.m
ADC_EXPERIMENT_9.m
ADC_EXPERIMENT_10.m
ADC_EXPERIMENT_16.m
/MATLAB Drive/ADC_EXPERIMENT_16.m
clc;
clear;
close all;
% Input Signal
t = 0:0.001:1;
x = sin(2*pi*5*t);
% Uniform Quantization (8 Levels)
L = 8;
x_uni = round((x+1)*(L-1)/2);
x_uni = (x_uni*2/(L-1))-1;
% Non-Uniform Quantization (μ-law)
mu = 255;
x_comp = sign(x).*log(1+mu*abs(x))/log(1+mu);
x_non = round((x_comp+1)*(L-1)/2);
x_non = (x_non*2/(L-1))-1;
% Plot Results
figure;
subplot(3,1,1)
Apps
Figure 1
Original Signal
Uniform Quantization
Non-Uniform Quantization (μ-law)
Command Window
>>
32°C
Mostly cloudy
Search
ENG
IN
13:27
14-10-2026

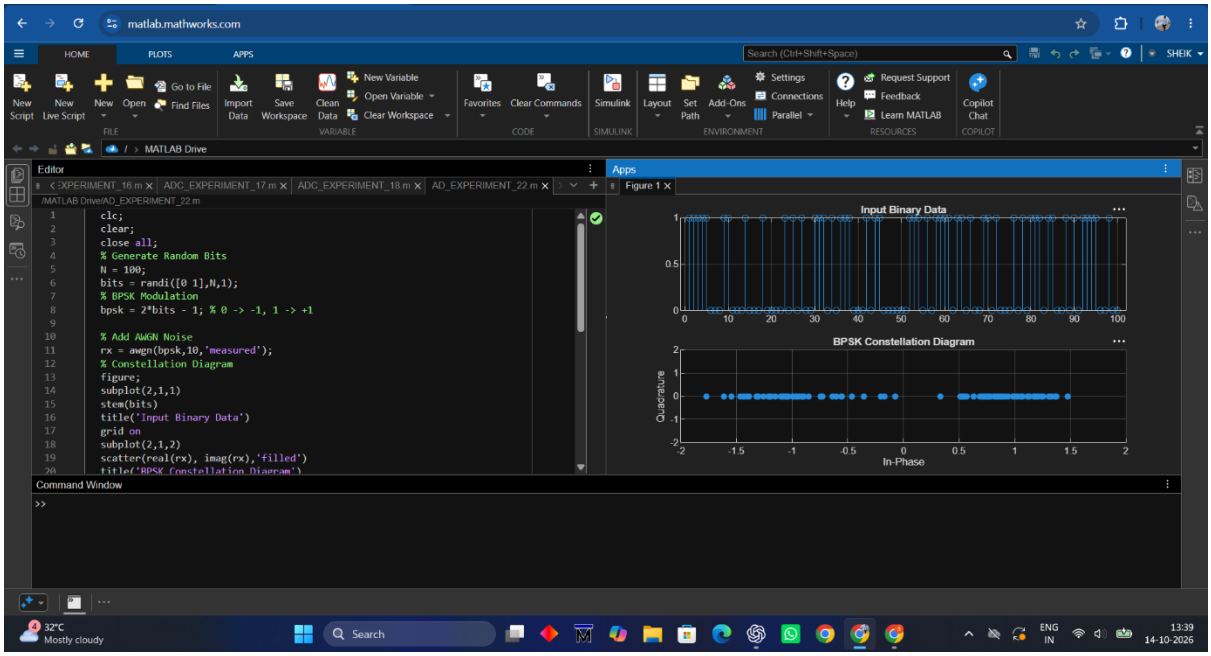
EXP 17


matlab.mathworks.com
HOME
PLOTS
APPS
MATLAB Drive
Editor
clc;
clear;
close all;
%% Parameters
N = 100; % Number of bits
sps = 20; % Samples per symbol
SNR = 10; % Signal-to-Noise Ratio (dB)
%% Generate Random Binary Data
data = randi([0 1],1,N);
%% NRZ Encoding (0 -> 0, 1 -> 1)
tx_signal = repelem(data,sps);
%% Pass Signal through AWGN Channel
rx_signal = awgn(tx_signal,SNR,'measured');
%% Time Axis
t = (0:length(tx_signal)-1)/sps;
%% Plot Results
figure;
% Original Binary Data
Command Window
Figure 2: Eye Diagram
Eye Diagram of Received Signal
Apps
Figure 1
Original Binary Data
Bit Number
Transmitted NRZ Signal
Received Signal with AWGN (SNR = 10 dB)
Time
32°C
Mostly cloudy
Search
ENG
IN
13:33
14-10-2026

EXP 18


matlab.mathworks.com
HOME
PLOTS
APPS
MATLAB Drive
Editor
clc;
clear;
close all;
%% Parameters
N = 100; % Number of bits
sps = 20; % Samples per symbol
%% Generate Random Binary Data
data = randi([0 1],1,N);
%% NRZ Encoding
tx = repelem(data,sps);
%% Time Axis
t = (0:length(tx)-1)/sps;
%% Introduce Inter-Symbol Interfere
% Simple Low-Pass FIR Filter (Chann
h = [0.2 0.6 1 0.6 0.2];
% Received Signal with ISI
rx = conv(tx,h,'same');
%% Plot Results%%
Command Window
Figure 2: Eye Diagram
X 0
Y 1.8
Amplitude
Time
Apps
Figure 1
Original Binary Data
Bit Number
Transmitted NRZ Signal
Received Signal with Inter-Symbol Interference (ISI)
32°C
Mostly cloudy
Search
ENG
IN
13:36
14-10-2026

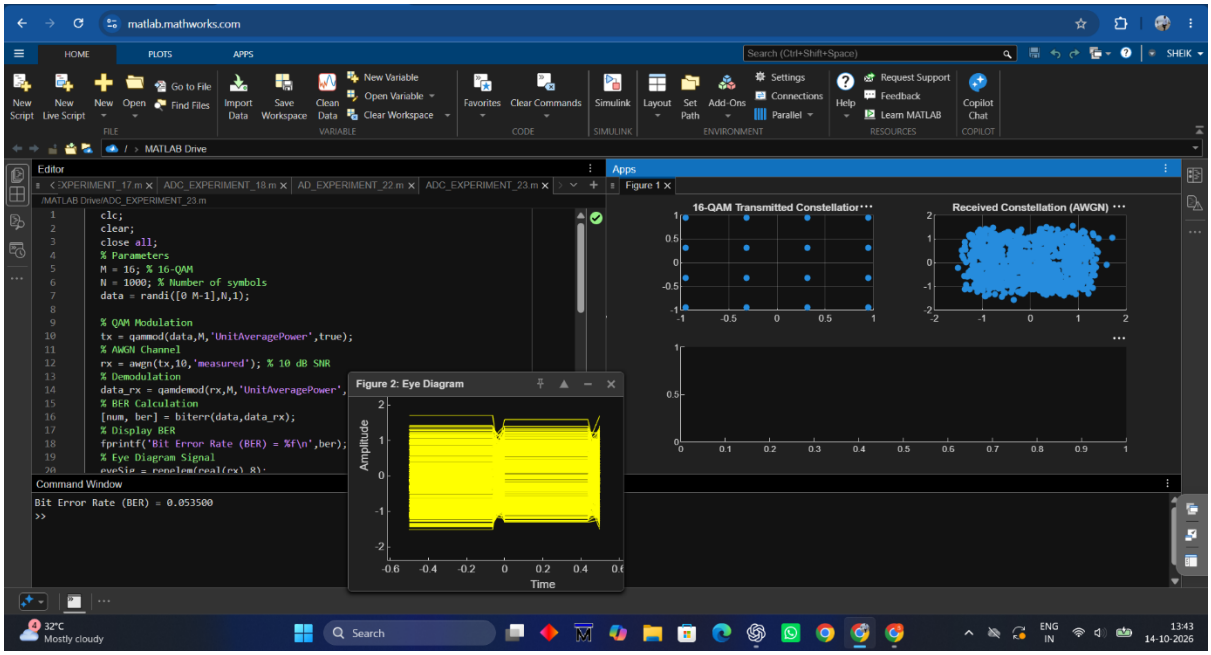
EXP 22




```
clc;
clear;
close all;
% Generate Random Bits
N = 100;
bits = randi([0 1],N,1);
% BPSK Modulation
bpsk = 2*bits - 1; % 0 -> -1, 1 -> +1

% Add AWGN Noise
rx = awgn(bpsk,10,'measured');
% Constellation Diagram
figure;
subplot(2,1,1)
stem(bits)
title('Input Binary Data')
grid on
subplot(2,1,2)
scatter(real(rx), imag(rx),'filled')
title('BPSK Constellation Diagram')
```

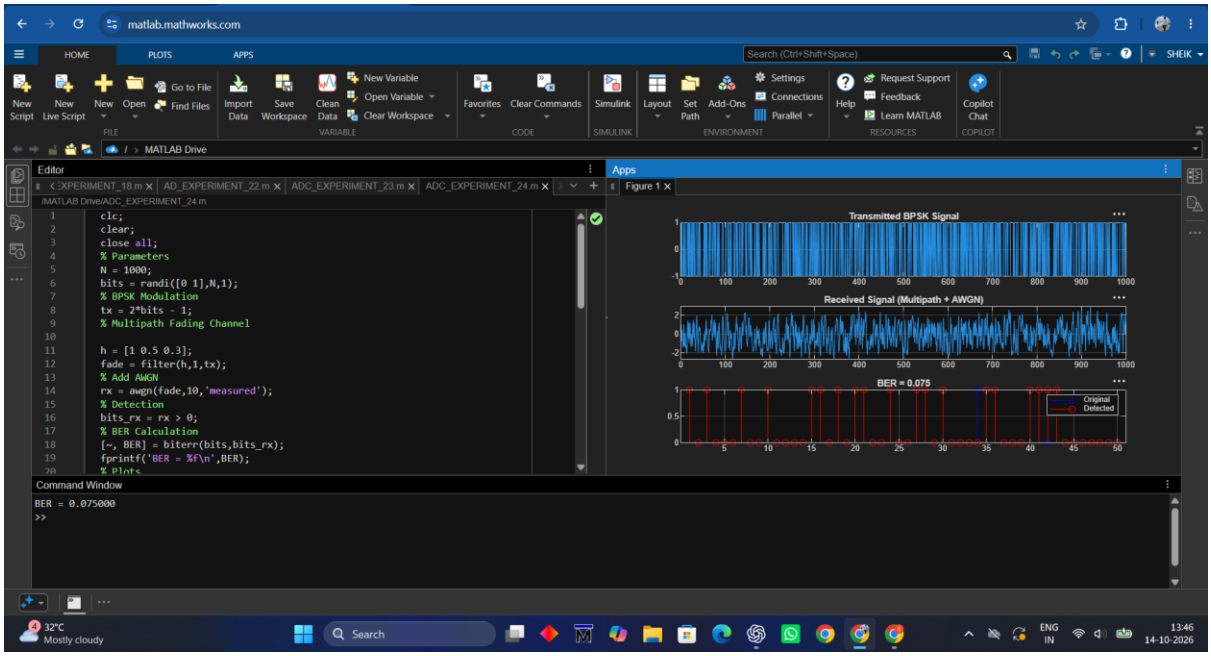
EXP 23



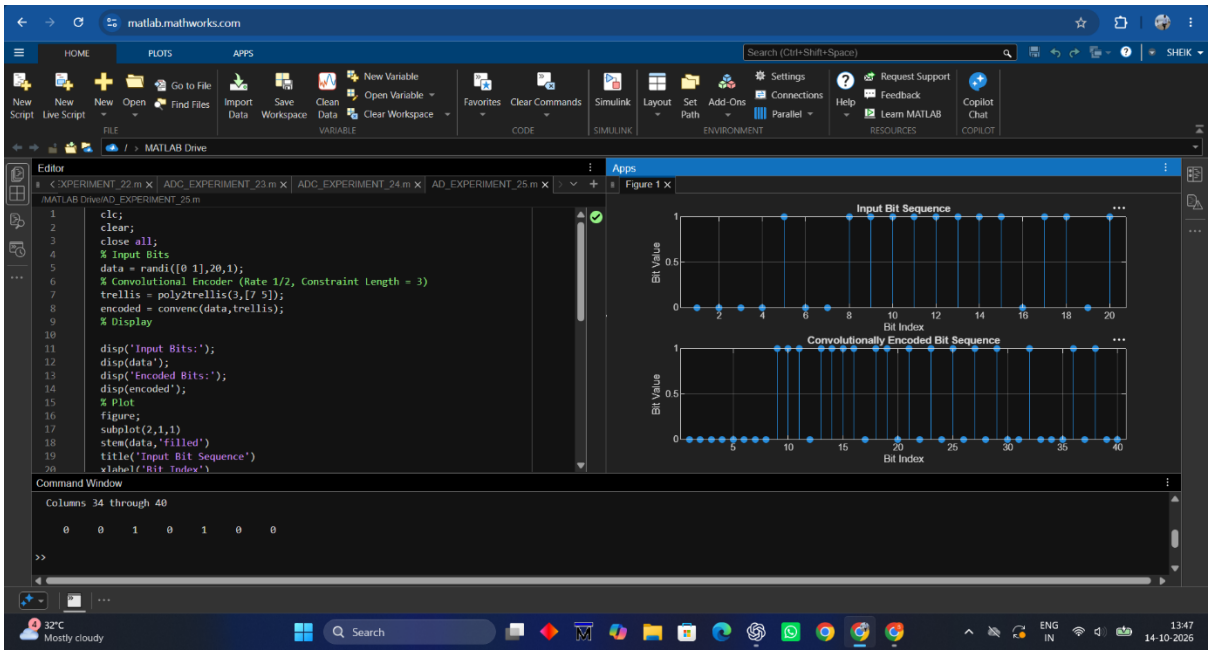

```
clc;
clear;
close all;
% Parameters
M = 16; % 16-QAM
N = 1000; % Number of symbols
data = randi([0 M-1],N,1);

% QAM Modulation
tx = qammod(data,M,'UnitAveragePower',true);
% AWGN Channel
rx = awgn(tx,10,'measured'); % 10 dB SNR
% Demodulation
data_rx = qamdemod(rx,M,'UnitAveragePower',
% BER Calculation
[num, ber] = biterr(data,data_rx);
% Display BER
fprintf('Bit Error Rate (BER) = %f\n',ber);
% Eye Diagram Signal
```

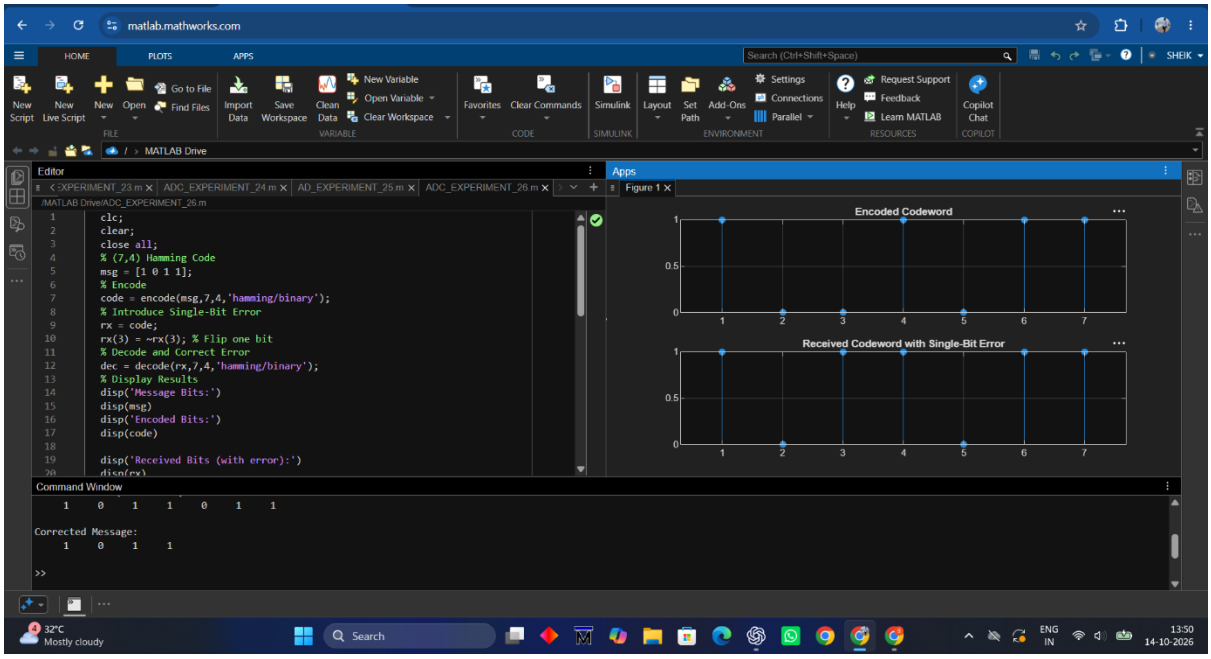
EXP 24


matlab.mathworks.com
HOME
PLOTS
APPS
Search (Ctrl+Shift+Space)
SHEIK
MATLAB Drive
Editor
Apps
Figure 1
Transmitted BPSK Signal
Received Signal (Multipath + AWGN)
BER = 0.075
Original
Detected
Command Window
BER = 0.075000
>>
32°C
Mostly cloudy
Search
ENG
IN
13:46
14-10-2026

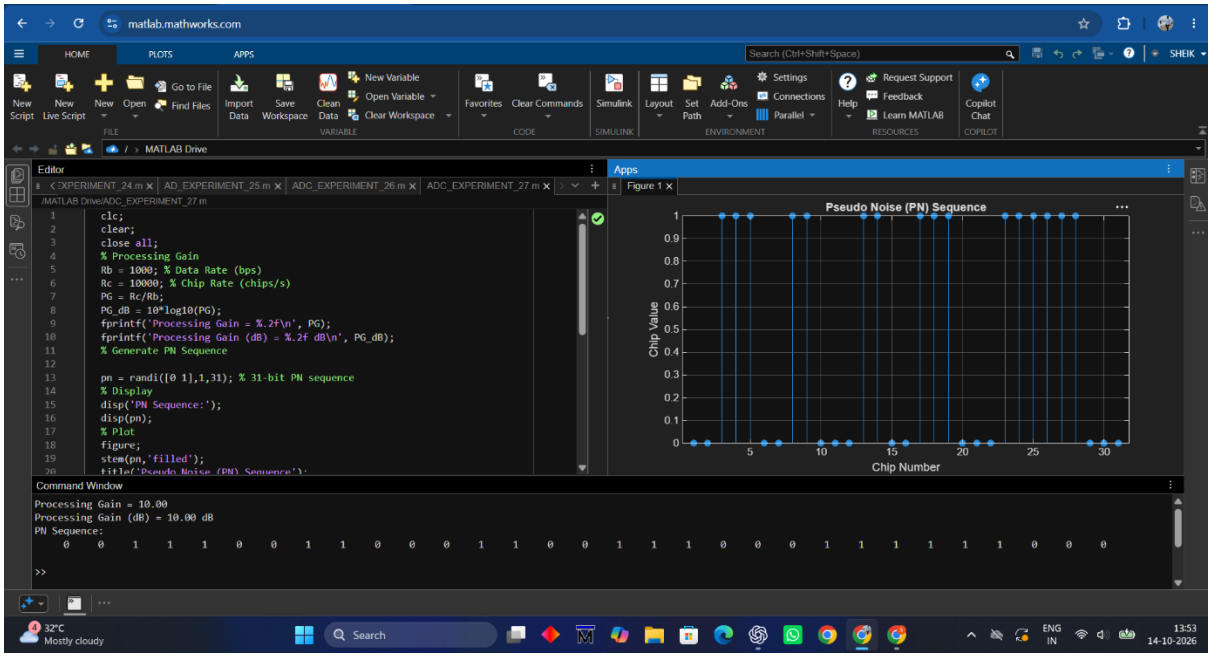
EXP 25


matlab.mathworks.com
HOME
PLOTS
APPS
Search (Ctrl+Shift+Space)
SHEIK
MATLAB Drive
Editor
Apps
Figure 1
Input Bit Sequence
Bit Value
Bit Index
Convolutionally Encoded Bit Sequence
Bit Value
Bit Index
Command Window
Columns 34 through 40
0 0 1 0 1 0 0
>>
32°C
Mostly cloudy
Search
ENG
IN
13:47
14-10-2026

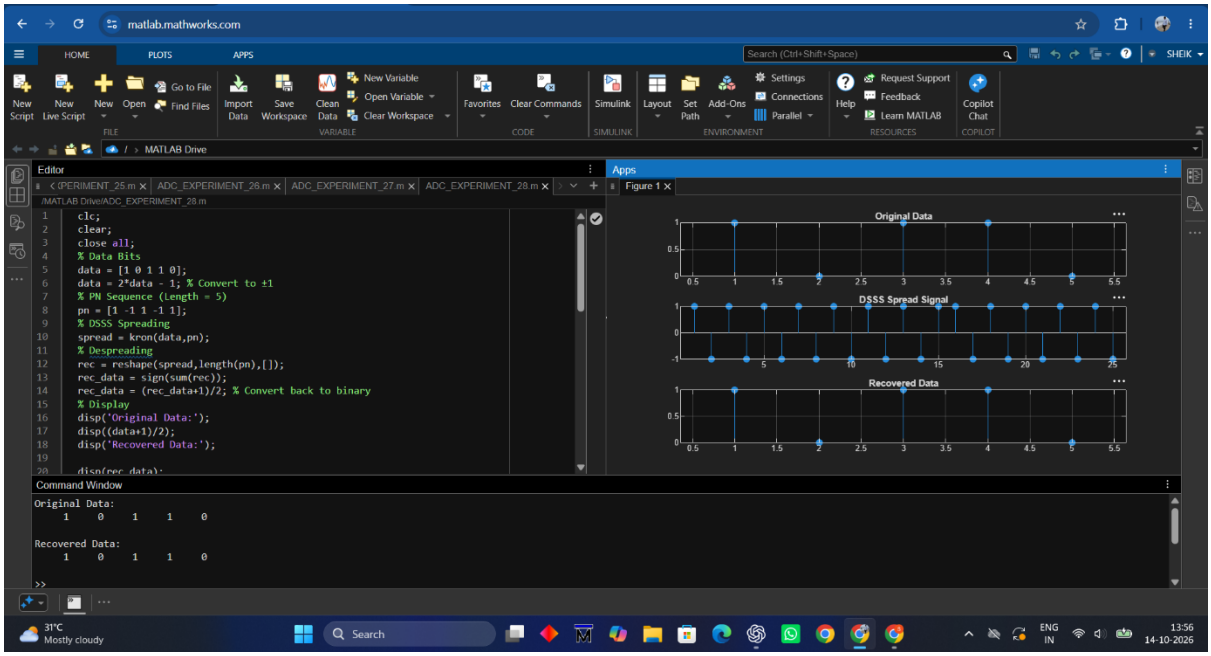
EXP 26


matlab.mathworks.com
HOME
PLOTS
APPS
Search (Ctrl+Shift+Space)
SHEIK
New Script
New Live Script
New
Open
Go to File
Find Files
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
Favorites
Clear Commands
Simulink
Layout
Set Path
Add-Ons
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SIMULINK
ENVIRONMENT
RESOURCES
COPILOT
MATLAB Drive
Editor
EXPERIMENT_23.m
ADC_EXPERIMENT_24.m
AD_EXPERIMENT_25.m
ADC_EXPERIMENT_26.m
/MATLAB Drive/ADC_EXPERIMENT_26.m
clc;
clear;
close all;
% (7,4) Hamming Code
msg = [1 0 1 1];
% Encode
code = encode(msg,7,4,'hamming/binary');
% Introduce Single-Bit Error
rx = code;
rx(3) = ~rx(3); % Flip one bit
% Decode and Correct Error
dec = decode(rx,7,4,'hamming/binary');
% Display Results
disp('Message Bits:')
disp(msg)
disp('Encoded Bits:')
disp(code)
disp('Received Bits (with error):')
Apps
Figure 1
Encoded Codeword
Received Codeword with Single-Bit Error
Command Window
1 0 1 1 0 1 1
Corrected Message:
1 0 1 1
>>
32°C
Mostly cloudy
Search
ENG
IN
13:50
14-10-2026

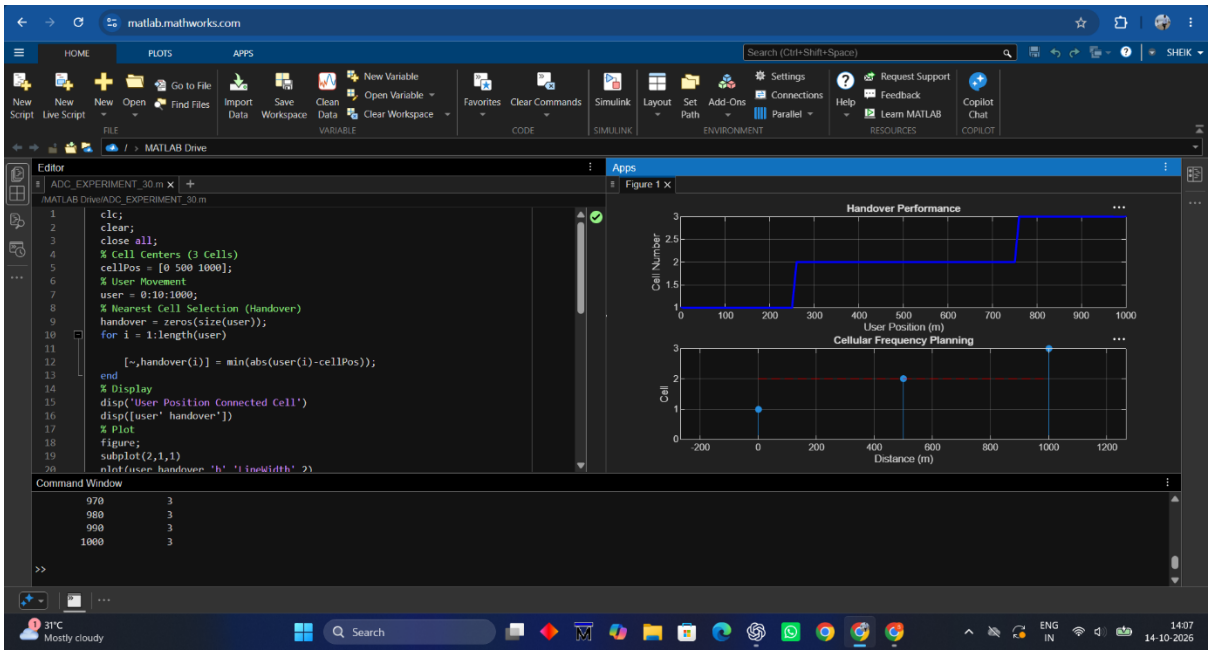
EXP 27


matlab.mathworks.com
HOME
PLOTS
APPS
Search (Ctrl+Shift+Space)
SHEIK
New Script
New Live Script
New
Open
Go to File
Find Files
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
Favorites
Clear Commands
Simulink
Layout
Set Path
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Connections
Parallel
Help
Request Support
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Learn MATLAB
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SIMULINK
ENVIRONMENT
RESOURCES
COPILOT
MATLAB Drive
Editor
EXPERIMENT_24.m
AD_EXPERIMENT_25.m
ADC_EXPERIMENT_26.m
ADC_EXPERIMENT_27.m
/MATLAB Drive/ADC_EXPERIMENT_27.m
clc;
clear;
close all;
% Processing Gain
Rb = 1000; % Data Rate (bps)
Rc = 10000; % Chip Rate (chips/s)
PG = Rc/Rb;
PG_dB = 10*log10(PG);
fprintf('Processing Gain = %.2f\n', PG);
fprintf('Processing Gain (dB) = %.2f dB\n', PG_dB);
% Generate PN Sequence
pn = randi([0 1],1,31); % 31-bit PN sequence
% Display
disp('PN Sequence:');
disp(pn);
% Plot
figure;
stem(pn,'filled');
Apps
Figure 1
Pseudo Noise (PN) Sequence
Chip Value
Chip Number
Command Window
Processing Gain = 10.00
Processing Gain (dB) = 10.00 dB
PN Sequence:
0 0 1 1 1 0 0 1 1 0 0 0 1 1 0 0 1 1 1 0 0 0 1 1 1 1 1 1 0 0 0
>>
32°C
Mostly cloudy
Search
ENG
IN
13:53
14-10-2026

EXP 28


matlab.mathworks.com
HOME
PLOTS
APPS
Search (Ctrl+Shift+Space)
SHEIK
MATLAB Drive
Editor
ADC_EXPERIMENT_28.m
clc;
clear;
close all;
% Data Bits
data = [1 0 1 1 0];
data = 2*data - 1; % Convert to ±1
% PN Sequence (Length = 5)
pn = [1 -1 1 -1 1];
% DSSS Spreading
spread = kron(data,pn);
% Despreading
rec = reshape(spread,length(pn),[]);
rec_data = sign(sum(rec));
rec_data = (rec_data+1)/2; % Convert back to binary
% Display
disp('Original Data:');
disp((data+1)/2);
disp('Recovered Data:');
Apps
Figure 1
Original Data
DSSS Spread Signal
Recovered Data
Command Window
Original Data:
1 0 1 1 0
Recovered Data:
1 0 1 1 0
31°C
Mostly cloudy
Search
ENG
IN
13:56
14-10-2026

EXP 30


matlab.mathworks.com
HOME
PLOTS
APPS
Search (Ctrl+Shift+Space)
SHEIK
MATLAB Drive
Editor
ADC_EXPERIMENT_30.m
clc;
clear;
close all;
% Cell Centers (3 Cells)
cellPos = [0 500 1000];
% User Movement
user = 0:10:1000;
% Nearest Cell Selection (Handover)
handover = zeros(size(user));
for i = 1:length(user)
[~,handover(i)] = min(abs(user(i)-cellPos));
end
% Display
disp('User Position Connected Cell')
disp([user' handover'])
% Plot
figure;
subplot(2,1,1)
Apps
Figure 1
Handover Performance
Cell Number
User Position (m)
Cellular Frequency Planning
Cell
Distance (m)
Command Window
970 3
980 3
990 3
1000 3
31°C
Mostly cloudy
Search
ENG
IN
14:07
14-10-2026